Probability Theory

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Preface

This repository contains teaching and learning materials prepared and used during "Introduction to biostatistics and machine learning" course, organized by NBIS, National Bioinformatics Infrastructure Sweden. The course is open for PhD students, postdoctoral researcher and other employees within Swedish universities. The materials are geared towards life scientists wanting to be able to understand and use basic statistical and machine learning methods. More about the course https://uppsala.instructure.com/courses/93277.

Learning outcomes

- understand the concept of random variables
- understand the concept of probability
- understand and learn to use resampling to compute probabilities
- understand the concept probability mass function
- understand the concept probability density function
- understand the concept cumulative distribution functions
- use normal distribution
- understand the central limit theorem
- to understand and define sampling distribution and standard error
- to compute standard error of mean and proportions

1 Introduction to probability

Some things are more likely to occur than others. Compare:

- the chance of the sun rising tomorrow with the chance that no-one is infected with COVID-19 tomorrow
- the chance of a dark winter in Stockholm with the chance of no rainy days over the summer months in Stockholm

We intuitively believe that the chance of sun rising or dark winter occurring are enormously higher than COVID-19 disappearing over night or having no rain over the entire summer. **Probability** gives us a scale for measuring the likeliness of events to occur. **Probability rules** enable us to reason about uncertain events. The probability rules are expressed in terms of sets, a well-defined collection of distinct objects.

Suppose we perform an experiment that we do not know the outcome of, i.e. we are uncertain about the outcome. We can however list all the outcomes that might occur.

- **sample space** is the set S of these possible outcomes of the experiment, e.g. getting 1, 2 etc. on the 6-sided dice $S = \{1, 2, 3, 4, 5, 6\}$
- **an event** is a subset of the sample space
- an event is said to **occur** if the outcome of the experiment belong to this set
- The **complement**, E', of the event E contains all the outcomes in S which are not in E
- Two sets E and F, such that $E \cap F = \emptyset$, are said to be **disjoint**

1.1 Axioms of probability

1. $0 \leq P(E) \leq 1$ for any event $E \subseteq S$
2. $P(S) = 1$
3. If E, F are disjoint events, then $P(E \cup F) = P(E) + P(F)$

1.2 Common rules of probability

Based on the axioms the following rules of probability can be proved.

- **Complement rule**: let $E \subseteq S$ be any event, then $P(E') = 1 - P(E)$

- **Impossible event**: $P(\emptyset) = 0$
- **Probability of a subset**: let $E, F \subseteq S$ be events such that $E \subseteq F$ then $P(F) \geq P(E)$
- **Addition rule**: let $E, F \subseteq S$ be any two events, then $P(E \cup F) = P(E) + P(F) - P(E \cap F)$

1.3 Conditional probability

Let $E, F \subseteq S$ be two events that $P(E) > 0$ then the conditional probability of F given that E occurs is defined to be:

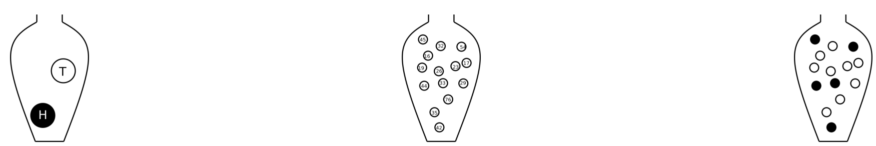
$$P(F|E) = \frac{P(E \cap F)}{P(E)}$$

Product rule follows conditional probability: let $E, F \subseteq S$ be events such that $P(E) > 0$ then:

$$P(E \cap F) = P(F|E)P(E) = P(E|F)P(F)$$

1.4 The urn model

The urn model is a simple mathematical model commonly used in statistics and probability. In the urn model, objects (such as people, mice, cells, genes, molecules, etc) are represented by balls of different colors or labels. A fair coin can be represented by an urn with two balls representing the coin's two sides; H and T. A group of people can be modeled in an urn model, if age is the variable of interest, we write the age of each person on the balls. If we instead are interested in if the people are allergic to pollen or not, we color the balls according to allergy status.



(a) Fair coin (b) Age (c) Pollen allergy status

Figure 1.1: Urn models of a fair coin, age of a group of people, and pollen allergy status of a group of people.

In the urn model every unit (ball) is equally likely of being selected. This means that the urn model is well suited to represent flipping a fair coin. However, a biased coin can also be modelled using an urn model, by changing the number of balls that represent each side of the coin.

By drawing balls from the urn with (or without) replacement, probabilities and other properties of the model can be inferred. For example, with an urn representinga population of people who are or are not allergic, the probability of randomly selecting three people who are all allergic can be determined.

1.5 Random variables

The outcome of a random experiment can be described by a **random variable**. Whenever chance is involved in the outcome of an experiment the outcome is a random variable.

A random variable can not be predicted exactly, but the probability of all possible outcomes can be described. The **sample space** is the set of all possible outcomes of a random variable. Note, the sample space is not always countable.

A random variable is usually denoted by a capital letter, $X, Y, Z, ...$ Values collected in an experiment are **observations** of the random variable, usually denoted by lowercase letters $x, y, z, ...$

The **sample space** is the collection of all possible observation values.

The **population** is the collection of all possible observations.

A **sample** is a subset of the population.

Example 1.1. Paint a fair six sided dice black on one side and white on the other five sides. The outcome of throwing this dice is a random variable, X. The possible outcomes, the sample space, is black and white. The population is all possible observations, i.e. the six sides.

Example random variables and probabilites:

- The weight of a random newborn baby, W. $P(W > 4.0kg)$
- The smoking status of a random mother, S. $P(S = 1)$
- The hemoglobin concentration in blood, Hb. $P(Hb < 125g/L)$
- The number of mutations in a gene
- BMI of a random man
- Weight status of a random man (underweight, normal weight, overweight, obese)
- The result of throwing a

Conditional probability can be written for example $P(W \geq 3.5|S = 1)$, which is the probability that $X \geq 3.5$ if $S = 1$, in words "the probability that a smoking mother has a baby with birth weight of 3.5 kg or more".

Random variables are divided based on their data type, categorical (nominal or ordinal) or numeric (discrete or continuous).

2 Discrete random variables

A categorical random variable has nominal or ordinal outcomes such as; {red, blue, green} or {tiny, small, average, large, huge}.

A discrete random number is usually counts and has a countable number of outcome values, such as {1,2,3,4,5,6}; {0,2,4,6,8} or all integers.

A discrete or categorical random variable can be described by its **probability mass function (PMF)**.

The probability that the random variable, X, takes the value x is denoted $P(X = x) = p(x)$. Note that:

1. $0 \leq p(x) \leq 1$, a probability is always between 0 and 1.
2. $\sum p(x) = 1$, the sum over all possible outcomes is 1.

Example 2.1 (The number of dots on a dice)**.** When rolling a dice the there are six possible outcomes; 1, 2, 3, 4, 5 and 6, each of which have the same probability, if the dice is fair. The outcome of one dice roll can be described by a random variable X. The probability of a particular outcome x is denoted $P(X = x)$ or $p(x)$.

The probability mass function of a fair six-sided dice can be summarized in a table;

Table 2.1: Probability mass function of a fair six-sided dice.

x	1	2	3	4	5	6
p(x)	0.167	0.167	0.167	0.167	0.167	0.167

or in a barplot;

Example 2.2 (Nucleotide in a given site)**.** The nucleotide at a given genomic site can be one of the four nucleotides; {A, C, T, G}. Unlike the sides on a dice the four nucleotides are usually not equally likely. The nucleotide at the site can be described by a random variable, X. The probability mass function can be summarized in a table or a bar plot.

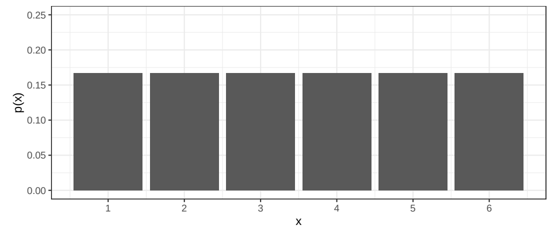


Figure 2.1: Probability mass function of a fair six-sided dice.

Table 2.2: Probability mass function of a nucleotide site.

x	A	C	T	G
p(x)	0.4	0.2	0.1	0.3

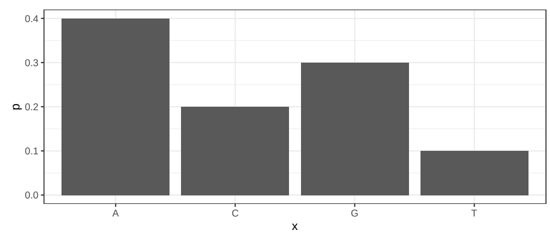


Figure 2.2: Probability mass function of a nucleotide site.

Example 2.3 (CFU)**.** The number of bacterial colony forming units (CFU) on a plate is a random number that can be described by a pmf such as in Figure 2.3.

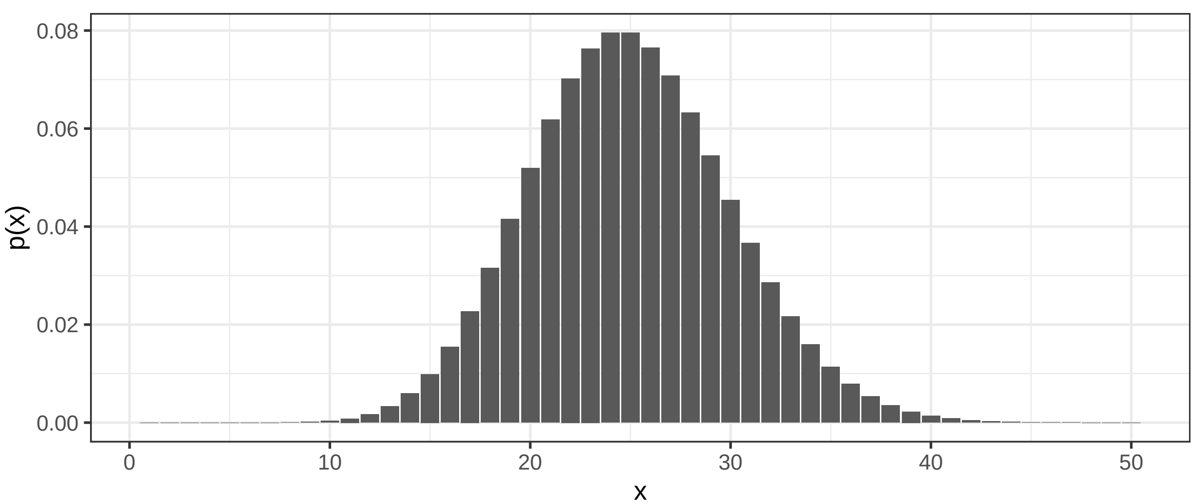


Figure 2.3: Probability mass distribution of the number of bacterial colonies on an agar plate.

The probability mass function can be used to compute the probability of an event, such as the events

- *get an even number when rolling a* - sum up the probabiliy for 2, 4, and 6, p(even)=p(2)+p(4)+p(6)=0.5
- *less than 15 CFUs on an agar plate* - sum upp the probabilities $P(X < 15) = \sum_{k=0}^{14} p(k) = 0.01$

The **cumulative distribution function (CDF)** is defined for numeric discrete random variables as $F(x) = P(X <= x)$.

2.1 Expected value and variance

Two common properties of a probability distribution, such as a probability mass function, are the **expected value** and the **variance**.

The **expected value** is the average outcome of a random variable over many trials and is denoted $E[X]$ or μ and can be computed by summing up all possible outcome values weighted by their probability;

$$E[X] = \mu = \sum_{i=1}^{n} x_i p(x_i),$$

where n is the number of outcomes.

For a **uniform distribution**, where every outcome has the same probability , the expected value can be computed as the sum of all outcome values divided by the total number of outcome values.

The expected value can also be computed as the population mean, i.e. by summing over outcome value for every object in the population (this is of course only possible if the population is countable);

$$E[X] = \mu = \frac{1}{N} \sum_{i=1}^{N} x_i,$$

where N is the number of objects in the population.

The **variance** is a measure of spread defined as the expected value of the squared difference of the random variable and its expected value;

$$var(X) = \sigma^2 = E[(X - E[X])^2] = \sum_{i=1}^{n} (x_i - \mu)^2 p(x_i).$$

The variance can also be computed by summing over the outcome value for every object in the population;

$$var(X) = \sigma^2 = \frac{1}{N} \sum_{i=1}^{N} (x_i - \mu)^2,$$

where N is the number of objects in the population.

The **standard deviation** is the square root of the **variance**. The standard deviation of a random variable is usually denoted σ. The standard deviation is always positive and on the same scale as the outcome values.

2.1.1 Linear transformations and combinations

$$E(aX) = aE(X)$$

$$E(X + Y) = E(X) + E(Y)$$

$$E[aX + bY] = aE[X] + bE[Y],$$

where a and b are constants.

$$var(aX) = a^2 var(X)$$

For independent random variables X and Y

$$var(aX + bY) = a^2 var(X) + b^2 var(Y)$$

2.2 Simulate distributions

Once a random variable's probability distribution is known, probabilities, such as $P(X = x), P(X < x)$ and $P(X \geq x)$, and properties, such as expected value and variance, of the random variable can be computed. If the distribution is not known, simulation might be the solution.

Example 2.4 (Simulate coin toss)**.** In a single coin toss the probabity of heads is 0.5. In 20 coin tosses, what is the probability of at least 15 heads?

The outcome of a single coin toss is a random variable, X, with two possible outcomes $\{H, T\}$. We know that $P(X = H) = 0.5$. The random variable of interest is the number of heads in 20 coin tosses, Y. The probability that we need to compute is $P(Y \geq 15)$.

A single coin toss can be modelled by an urn with two balls. When a ball is drawn randomly from the urn, the probability to get the black ball (heads) is $P(X = H) = 0.5$.

In R we can simulate random draws from an urn model using the function `sample`.

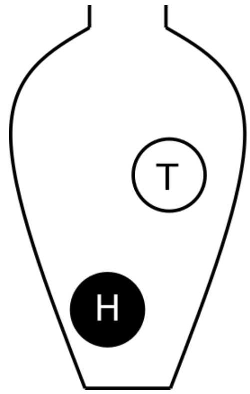


Figure 2.4: A coin toss. Urn model with one black ball (heads) and one white ball (tails).

```
## A single coin toss
sample(c("H", "T"), size=1)
```

```
[1] "T"
```

```
## Another coin toss
sample(c("H", "T"), size=1)
```

```
[1] "T"
```

Every time you run `sample` a new coin toss is simulated.

If we want to simulate tossing 20 coins (or one coin 20 times) we can use the same urn model, if the ball is replaced after each draw.

The argument `size` tells the function how many balls we want to draw from the urn. To draw 20 balls from the urn, set `size=20,` remember to replace the ball after each draw!

```
## 20 independent coin tosses
(coins <- sample(c("H", "T"), size=20, replace=TRUE))
```

```
 [1] "T" "H" "T" "H" "T" "T" "H" "H" "H" "H" "T" "H" "H" "T" "T" "H" "H" "T" "T"
[20] "T"
```

How many heads did we get in the 20 random draws?

```r
## How many heads?
sum(coins == "H")
```

```
[1] 10
```

We can repeat this experiment (toss 20 coins and count the number of heads) several times to estimate the distribution of number of heads in 20 coin tosses.

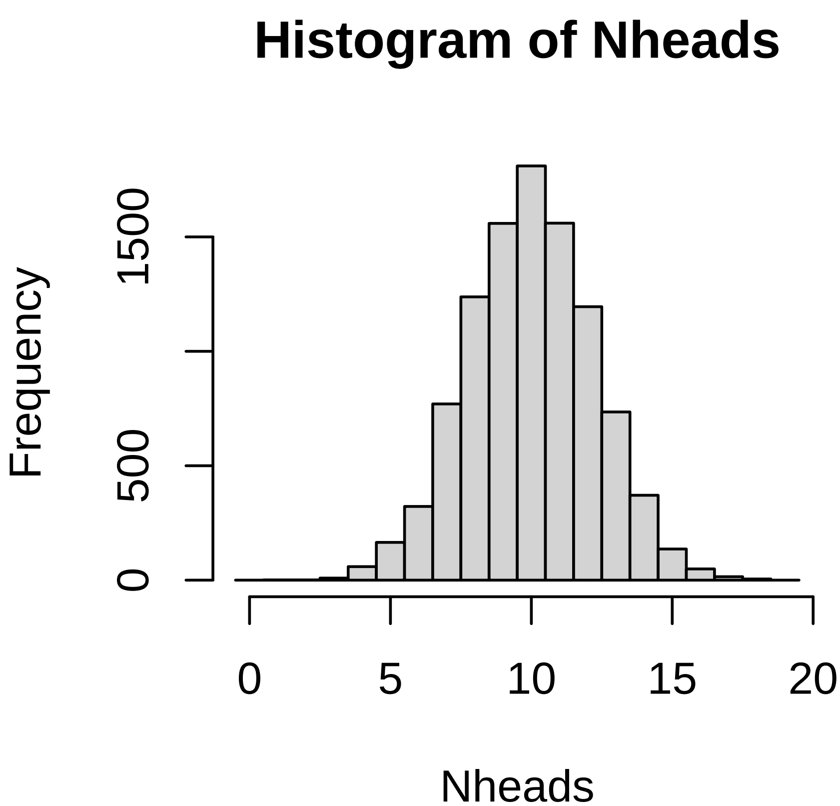
To do the same thing several times we use the function `replicate`.

To simulate tossing 20 coins and counting the number of heads 10000 times, do the following;

```r
Nheads <- replicate(10000, {
  coins <- sample(c("H", "T"), size=20, replace=TRUE)
  sum(coins == "H")
})
```

Plot the distribution of the number of heads in a histogram.

```r
hist(Nheads, breaks=0:20-0.5)
```



Now, let's get back to the question; when tossing 20 coins, what is the probability of at least 15 heads?

$P(Y \geq 15)$

Count how many times out of our 10000 experiments the number is 15 or greater

```
sum(Nheads >= 15)
```

```
[1] 205
```

From this we conclude that

$P(Y \geq 15) = 205/10000 = 0.0205$

Resampling can also be used to compute other properties of a random variable, such as the expected value.

The **law of large numbers** states that if the same experiment is performed many times the average of the result will be close to the expected value.

The coin flip is a common example in statistics, but many situations with two outcomes can be modeled using similar models. If we consider the outcome of interest succes (like heads in the coin example) and the alternative outcome failure we can for example model;

- Drug effect: A patient can respond to drug treatment (success) or not (failure)
- Side effect: After a treatment a patient might experience a side effect (success) or not (failure)
- Treatment or placebo: Randomly assign a study participant into treatment or placebo group
- Antibiotic resistance: A bacteria is either resistant to an antibiotic or not

The probability of success and failure can be equal, like in the coin example, but they don't have to be equal. If the probability of a patient responding to a treatment is $p = 0.80$, we know that the probability of the patient not responding is $1 - p = 0.20$, this can be modelled using an urn model with 4 black balls (cured) and 1 white ball (not cured).

2.3 Parametric discrete distributions

A discrete parametric distribution is a probability distribution described by a set of parameters. In many situations data can be assumed to follow a parametric distribution.

2.3.1 Uniform

In a uniform distribution every possible outcome has the same probability. With n different outcomes, the probability for each outcome is $1/n$.

2.3.2 Bernoulli

A Bernoulli trial is a random experiment with two outcomes; success and failure. The probability of success, $P(success) = p$, is constant. The probability of failure is $P(failure) = 1 - p$.

When coding it is convenient to code success as 1 and failure as 0.

The outcome of a Bernoulli trial is a discrete random variable, X.

$$P(X = x) = p(x) = \begin{cases} p & \text{if } x = 1, success \\ 1 - p & \text{if } x = 0, failure \end{cases}$$

Using the definitions of expected value and variance it can be shown that;

$$E[X] = p$$
$$var(X) = p(1 - p)$$

A Bernoulli trial can be simulated, as seen in previous section, using the function `sample`.

2.3.3 Binomial

The number of successes in a series of independent and identical Bernoulli trials is a **binomial** random variable, X.

$X = \sum_{i=0}^{n} Z_i$,

where all Z_i describe the outcome of independent and identical Bernoulli trials with probability p for *success* ($P(Z_i = 1) = p$).

The probability mass function of X is called the binomial distribution. In short we use the notation;

$$X \sim Bin(n, p)$$

The probability mass function is

$$P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}$$

It can be shown that

$$E[X] = np$$
$$var(X) = np(1-p)$$

A binomial random variable is the number of successes when sampling n objects *with* replacement from an urn with objects of two types, of which the interesting type (success) has probability p.

The probability mass function, $P(X = k)$ can be computed using the R function `dbinom` and the cumulative distribution function $P(X \leq k)$ can be computed using `pbinom`.

Examples fo binomial random variables;

- The number of patients responding to a treatment out of n patients in a study, if the probability of a patient responding to treatment is p.
- The number of patients experiencing a side effect out of n patients in a study, if the probability of a side effect is p.
- the number of mutations in a gene of length n, if the mutations are independent and identically distributed and the probability of a mutation at every single position is p.

2.3.4 Poisson

The Poisson distribution describe the number of times a rare event occurs in a large number of trials. Commonly used to describe the number of events during a given time period.

A rare disease has a very low probability for a single individual. The number of individuals in a large population that catch the disease in a certain time period can be modelled using the Poisson distribution.

The probability mass function has a single parameter, λ, the expected value, and can be described as;

$$P(X = k) = \frac{\lambda^k}{k!} e^{-\lambda}$$

The expected value $\lambda = n\pi$, where n is the number of objects sampled from the population and π is the probability of a single object.

The variance is equal to the expected value;

$$var(X) = E[X] = \lambda = n\pi$$

The Poisson distribution can approximate the binomial distribution if n is large ($n > 20$) and π is small ($\pi < 0.05$ and $n\pi < 10$).

Examples of Poisson random variables;

- A rare disease has a very low probability for a single individual. The number of individuals in a large population that catch the disease in a certain time period is a Poisson random variable.
- Number of reads aligned to a gene region

2.3.5 Negative binomial

A negative binomial distribution describes the number of failures that occur before a specified number of successes (r) has occurred, in a sequence of independent and identically distributed Bernoilli trials. r is also called the dispersion parameter.

In R: `dnbinm, pnbinom, qnbinom`

2.3.6 Geometric

The geometric distribution is a special case of the negative binomial distribution, where $r = 1$.

In R: `dgeom, pgeom, qgeom`

2.3.7 Hypergeometric distribution

The hypergeometric distribution occurs when sampling n objects *without* replacement from an urn with N objects of two types, of which the interesting type has probability p.

The probability mass function

$$P(X = k) = \frac{\binom{Np}{k}\binom{N-Np}{n-k}}{\binom{N}{n}}$$

can be computed in R using `dhyper` and the cumulative distribution function $P(X \leq k)$ can be computed using `phyper`.

Examples of hypergeometric random variables;

- In a student group of 25 individuals, 10 are R beginners. If 5 individuals are randomly choosen to belong to group A, the number of R beginners in group A is a hypergeometric random variable.
- A drug company is producing 1000 pills per day, 5% have an amount of active substance below an acceptable threshold. In a random sample of 10 pills, how many contain to little active substance? The number of pills with to little active substance is a hypergeometric random variable.
- You investigate the differential expression of 10000 genes, 150 of these genes belong to the super interesting pathway XXX. You run a black-box algorithm that report 80 differentially expressed (DE) genes. If the black-box algorithm choose DE genes at random, the number of DE genes that belong to pathway XXX is a hypergeometric random variable.

2.3.8 Summary of discrete distributions

The binomial, hypoergeometric, negative binomial and poisson distributions have similarities. For large N (large population) the hypergeometric and binomial distributions are very similar. The Poisson distribution has equal variance and mean, whereas the binomial has a variance less than the mean and the negative binomial has a variance greater than the mean.

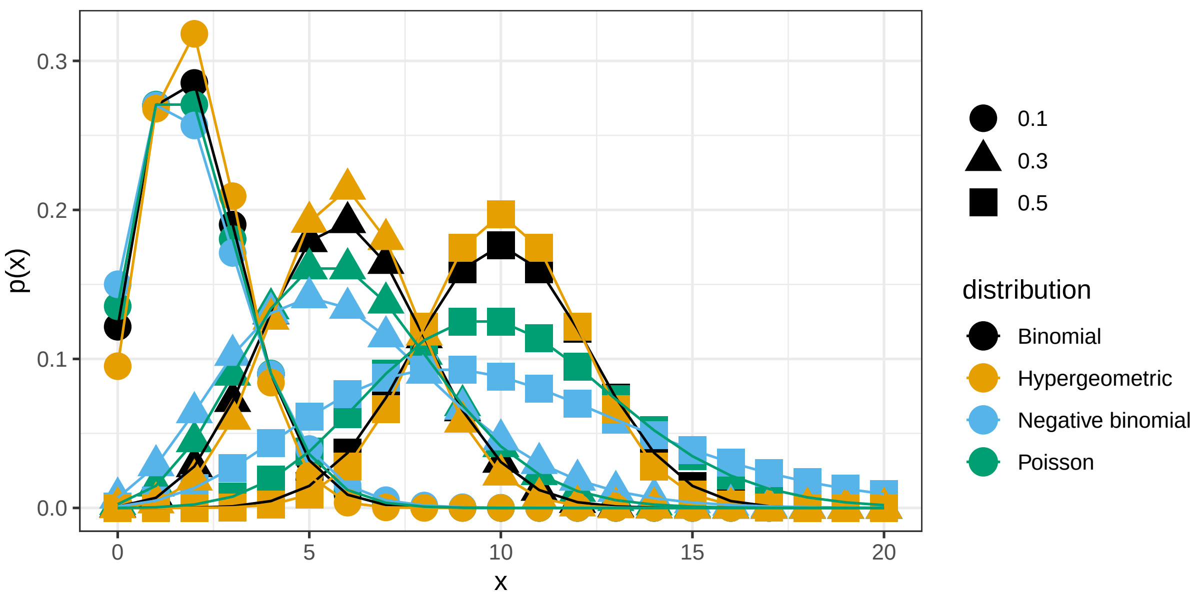


Figure 2.5: Probability mass functions for the binomial distribution (n=20, p=0.1, 0.3 or 0.5), hypergeometric distribution (N=100, n=20, p=0.1, 0.3 or 0.5), negative binomial distribution (n=20, r=n*p, p=0.1, 0.3 or 0.5) and Poisson distribution (n=20, p=0.1, 0.3 or 0.5).

Probability mass functions, $P(X = x)$, for the binomial, hypergeometric, negative binomial and Poisson distributions can in R can be computed using functions `dbinom`, `dhyper`, `dnbinom` and `dpois`, respectively.

Cumulative distribution functions, $P(X \leq x)$ can be computed using `pbinom`, `phyper`, `pnbinom` and `ppois`.

Also, functions for computing an x such that $P(X \leq x) = q$, where q is a probability of interest are available using `qbinom`, `qhyper`, `qnbinom` and `qpois`.

Exercises: Discrete random variables

Introduction to probability

Exercise 2.1 (BRCA)**.** The probability of carrying mutations (one or more) in the breast cancer gene BRCA1 is 0.01. What is the probability of not carrying any mutations in BRCA1?

Hint

Use the rule of complement.

Solution

According to the rule of complement $P(no\,mutation) = 1 - P(mutations) = 1 - 0.01 = 0.99$

Exercise 2.2 (A coin toss)**.** When tossing a fair coin

a) what is the probability of heads?
b) what is the probability of tails?

Hint

Fair = equal probabilities

Solution

For a fair coin the probability of heads and tails are equal; $P(H) = P(T)$. According to the rule of complement $p(H) = 1 - P(T)$
It follows that

a) $P(H) = 0.5$
b) $P(T) = 0.5$

Exercise 2.3 (Number of children)**.** In a region in Sweden with many children the number of children per household is between 0 and 6. The probability mass function is as follows;

x	0	1	2	3	4	5	6
p(x)	0.14	0.20	0.27	0.19	0.13	0.05	0.02

In a randomly choosen household

a) what is the probability of exactly 3 children?
b) what is the probability of less than 3 children?
c) what is the probability of 3 or less children?
d) what is the probability of an even number of children?

In your answers, denote the probability with a mathematical expression (such as $P(X > 4)$) and calculate its value.

Solution

The number of children in a random household is a random variable. Let X (a random variable) denote the number of children in a household in the studied region.

a) $P(X = 3) = 0.19$
b) $P(X < 3) = P(X = 0) + P(X = 1) + P(X = 2) = 0.14 + 0.20 + 0.27 = 0.61$
c) $P(X \leq 3) = P(X = 3) + P(X < 3) = 0.19 + 0.61 = 0.80$
d) $P(even\, X) = P(X = 0) + P(X = 2) + P(X = 4) + P(X = 6) = 0.14 + 0.27 + 0.13 + 0.02 = 0.56$

Exercise 2.4 (Rolling dice)**.** When tossing a fair six-sided dice

a) what is the probability of getting 6?
b) what is the probability of an even number?
c) what is the probability of getting 3 or more?
d) what is the expected value of dots on the dice?

Hint

On fair sided dice, all six sides have equal pobability.

Solution

The random variable, X, describe the number of dots on the upper face of a dice.

a) $P(X = 6) = \frac{1}{6}$
b) $P(even\, X) = \frac{3}{6} = \frac{1}{2}$
c) $P(X \geq 3) = \frac{4}{6} = \frac{2}{3}$

d) $E[X] = 1 * \frac{1}{6} + 2 * \frac{1}{6} + 3 * \frac{1}{6} + 4 * \frac{1}{6} + 5 * \frac{1}{6} + 6 * \frac{1}{6} = 3.5$

Simulation

Exercise 2.5 (Randomization)**.** In a clinical trial, enrolled patients are randomly assigned to treatment or control group with equal probability.

For a single patient, what is the probability of being assigned to

a) the treatment group?
b) the control group?

If 20 patients are enrolled in the study;

c) what is the probability of exactly 15 in the treatment group?
d) what is the probability of less than 7 in the treatment group?
e) What is the most probable number of patients in the treatment group?
f) what is the probability of 5 or less patients in the control group?
g) what is the probability of 2 or less patients in the treatment group?

Answer

a. $P(T) = 0.5$
b. $P(C) = 0.5$

Solution

a. The probability of assigning to control (C) and treatment (T) group are equal and sum up to 1. Hence $P(T) = 0.5$ and;

b. $P(C) = 0.5$

Simulate the assignment of patients into T or C groups.

```
## Randomization for a single patient
sample(c("T", "C"), size=1)
```

```
[1] "T"
```

```
## Randomize 20 independent patients
(patients <- sample(c("T", "c"), size=20, replace=TRUE))
```

```
 [1] "C" "T" "T" "C" "C" "T" "T" "T" "T" "T" "T" "T" "C" "T" "T" "C" "C" "C" "T"
[20] "C"
```

```
## How many patients are assigned to treatment group?
sum(patients == "T")
```

```
[1] 12
```

```
## Simulate by repeating 10000 times
Ntreat <- replicate(10000, {
  patients <- sample(c("T", "C"), size=20, replace=TRUE)
  sum(patients == "T")
})
```

c. Probability of exactly 15 T

```
## Proportion of the 10000 repeats with exactly 15 T
mean(Ntreat==15)
```

```
[1] 0.014
```

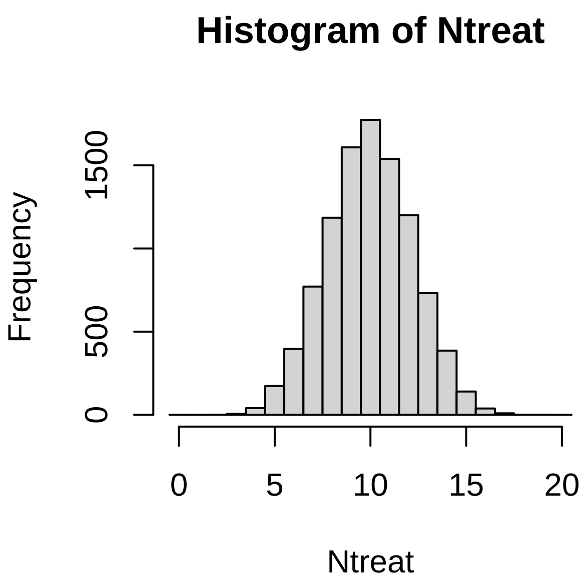
d. Probability of less than 7 T

```
mean(Ntreat<7)
```

```
[1] 0.062
```

e. What is the most probable number of T patients?

```
## plot the distribution and read the graph
hist(Ntreat, breaks=0:21-0.5)
```




```
## or tabulate
table(Ntreat)
```

```
Ntreat
   2    3    4    5    6    7    8    9   10   11   12   13   14   15   16   17
   1    6   40  173  397  771 1185 1608 1773 1539 1200  732  386  140   38    9
  18   19
   1    1
```

f. What is the probability of 5 C or less?

To get five or less C out of 20 throws is equal to getting 15 or more T out of 20.

```
## probability of 15 T or more
mean(Ntreat>=15)
```

```
[1] 0.019
```

g. what is the probability of 2 T or less?

```
mean(Ntreat<=2)
```

```
[1] 1e-04
```

```
sum(Ntreat<=2)
```

```
[1] 1
```

```r
## with this low number of observations, more repeats is required to get a more accurate an

Ntreat <- replicate(1000000, {
  patients <- sample(c("T", "C"), size=20, replace=TRUE)
  sum(patients == "T")
})
sum(Ntreat<=2)
```

```
[1] 210
```

```r
mean(Ntreat<=2)
```

```
[1] 0.00021
```

Exercise 2.6 (Bacterial colonies)**.** In a bacterial sample, 1/6 are antibiotic resistant. From bacterial colonies on an agar plate, you randomly pick 10 colonies and investigate how many that are antibiotic resistant.

a) Define the random variable of interest
b) What are the possible outcomes?
c) Using simulation, estimate the probability mass function
d) what is the probability to get at least 5 antibiotic resistant colonies?
e) Which is the most likely number of antibioitic colonies?
f) What is the probability to get exactly 2 antibiotic resistant colonies?
g) On average how many antibiotic resistant colonies would you get if the experiment is repeated many time?

Hint

Think of the dice example, where the probability of getting 'six' on one dice is 1/6.

Solution

a. X, the number of antibiotic resistant colonies out of 10.

b. $0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10$

c.

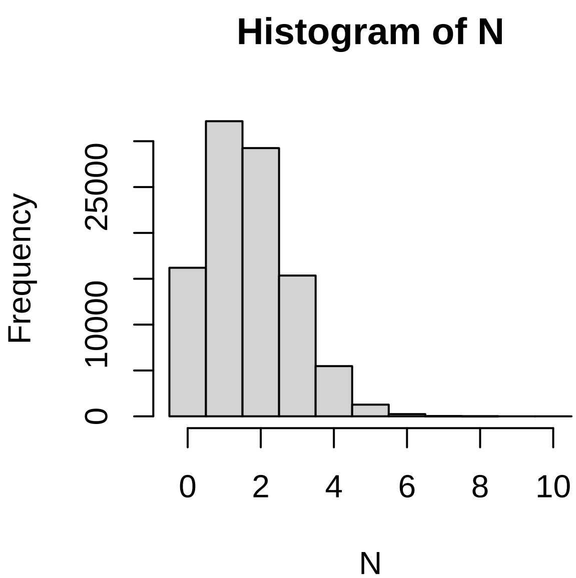
```r
##Simulate picking 10 colonies and counting the number of antibiotic resistant ones.
N <- replicate(100000, sum(sample(1:6, size=10, replace=TRUE)==6))
table(N)
```

```
N
    0     1     2     3     4     5     6     7     8
16200 32180 29251 15351  5474  1275   242    25     2
```

```r
##The probability mass function
table(N)/length(N)
```

```
N
      0       1       2       3       4       5       6       7       8
0.16200 0.32180 0.29251 0.15351 0.05474 0.01275 0.00242 0.00025 0.00002
```

```r
hist(N, breaks=(0:11)-0.5)
```



d) 0.015

```
[1] 1544
```

```
[1] 0.015
```

```
[1] 0.015
```

e) 1 (Use the PMF to answer this question)

f) 0.29

```r
mean(N==2)
```

```
[1] 0.29
```

g) 1.7

```r
mean(N)
```

```
[1] 1.7
```

```r
10*1/6
```

```
[1] 1.7
```

Exercise 2.7 (Pollen allergy)**.**

a. 30% of a large population is allergic to pollen. If you randomly select 3 people to participate in your study, what is the probability than none of them will be allergic to pollen?

Solution

```r
## Solution using 100 replicates
x <- replicate(100, sum(sample(c(0,0,0,0,0,0,0,1,1,1), size=3, replace=TRUE)))
table(x)
```

```
x
 0  1  2  3 
31 42 26  1 
```

```r
mean(x==0)
```

```
[1] 0.31
```

```r
## Solution using 1000 replicates
x <- replicate(1000, sum(sample(c(0,0,0,0,0,0,0,1,1,1), size=3, replace=TRUE)))
table(x)
```

```
x
```

```
  0   1   2   3 
355 418 195  32 
```

```r
mean(x==0)
```

```
[1] 0.35
```

```r
## Solution using 100000 replicates
x <- replicate(100000, sum(sample(c(0,0,0,0,0,0,0,1,1,1), size=3, replace=TRUE)))
table(x)
```

```
x
    0     1     2     3 
34146 44183 18933  2738 
```

```r
mean(x==0)
```

```
[1] 0.34
```

b. In a class of 20 students, 6 are allergic to pollen. If you randomly select 3 of the students to participate in your study, what is the probability than none of them will be allergic to pollen?

Solution

```r
## Solution using 100000 replicates
x <- replicate(100000, sum(sample(rep(c(0, 1), c(14, 6)), size=3, replace=FALSE)))
table(x)
```

```
x
    0     1     2     3 
31935 48080 18234  1751 
```

```r
mean(x==0)
```

```
[1] 0.32
```

c. Of the 200 persons working at a company, 60 are allergic to pollen. If you randomly select 3 people to participate in your study, what is the probability that none of them are allergic to pollen?

Solution

```
## Solution using 100000 replicates
x <- replicate(100000, sum(sample(rep(c(0, 1), c(140, 60)), size=3, replace=FALSE)))
table(x)
```

```
x
    0     1     2     3 
34180 44367 18943  2510 
```

```
mean(x==0)
```

```
[1] 0.34
```

d. Compare your results in a, b and c. Did you get the same results? Why/why not?

Solution

The results differ. In a the probability of selecting an allergic person is constant, regardless of the status of previously selected persons. On the other hand in the situations in b and c, the probability of selecting an allergic person changes depending on the persons selected before.

Parametric discrete distribution

Exercise 2.8 (Pollen)**.** Do Exercise 2.7 again, but using parametric distributions. Compare your results.

Solution

```
## 1.6 Solution using the Binomial distribution
pbinom(0, 3, 0.3)
```

```
[1] 0.34
```

```
## 1.7 Solution using the hypergeometric distribution
phyper(0, 6, 20-6, 3)
```

```
[1] 0.32
```

```r
## 1.8 Solution using the hypergeometric distribution
phyper(0, 60, 200-60, 3)
```

```
[1] 0.34
```

Exercise 2.9 (Gene set enrichment analysis)**.** You have analyzed 20000 genes and a bioinformatician you are collaborating with has sent you a list of 1000 genes that she says are important. You are interested in a particular pathway A. 200 genes in pathway A are represented among the 20000 genes, 20 of these are in the bioinformaticians important list.

If the bioinformatician selected the 1000 genes at random, what is the probability to see 20 or more genes from pathway A in this list?

Solution

```r
phyper(20, 200, 20000-200, 1000, lower.tail=FALSE)
```

```
[1] 0.0011
```

Exercise 2.10 (Chance of meeting boss)**.** Your boss comes in to the office three days per week. You do also come in to work three days per week. If you both choose which days to come in to work at random, what is the probability that a particular week you are in the office at the same time 0, 1, 2 or 3 days, respectively?

Solution

```r
x <- replicate(100000, {x<-sample(1:5, 3); y<-sample(1:5,3); length(intersect(x,y))})
table(x)
```

```
x
    1     2     3 
29892 60158  9950 
```

```r
dhyper(0:3, 3, 2, 3)
```

```
[1] 0.0 0.3 0.6 0.1
```

Exercise 2.11 (Rare disease)**.** A rare disease affects 3 in 100000 in a large population. If 10000 people are randomly selected from the population, what is the probability

a) that no one in the sample is affected?

b) that at least two in the sample are affected?

Solution

a)

```
n <- 10000
p <- 3/100000
ppois(0, n*p)
```

```
[1] 0.74
```

b)

```
ppois(1, n*p, lower.tail=FALSE)
```

```
[1] 0.037
```

3 Continuous random variable

A continuous random number is not limited to discrete values, but any continuous number within one or several ranges is possible.

Examples: weight, height, speed, intensity, …

A continuous random variable can be described by its **probability density function**, pdf.

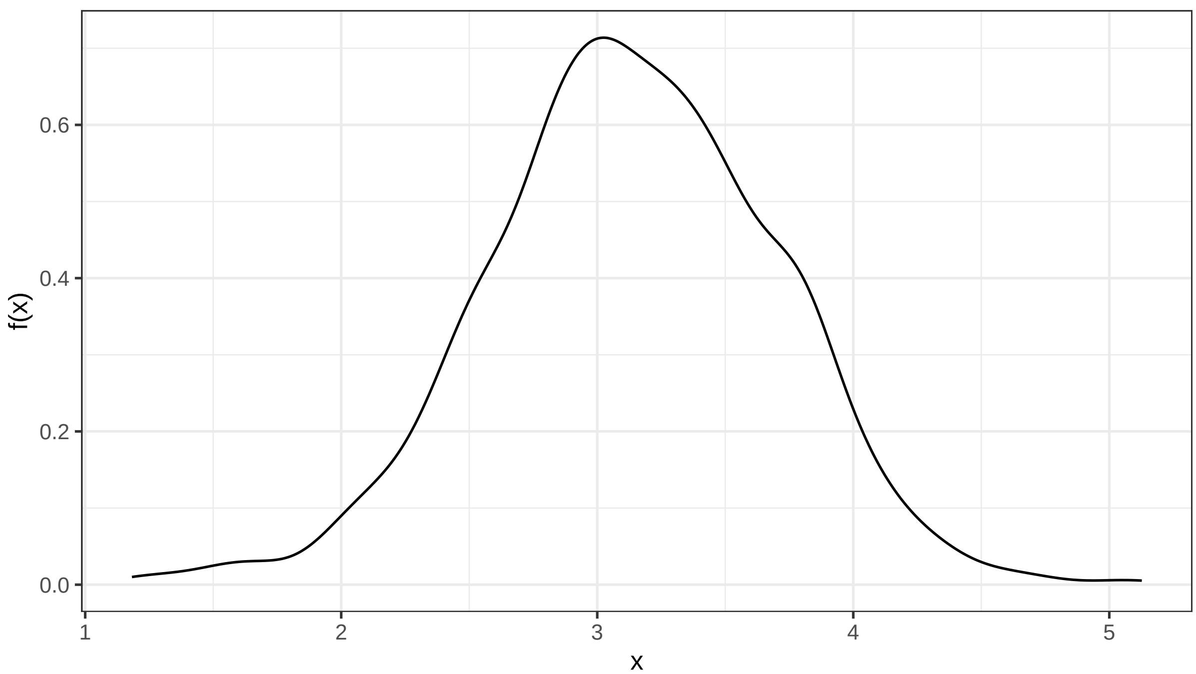


Figure 3.1: Probability density function of the weight of a newborn baby.

The probability density function, $f(x)$, is defined such that the total area under the curve is 1.

$$\int_{-\infty}^{\infty} f(x)dx = 1$$

The area under the curve from a to b is the probability that the random variable X takes a value between a and b.

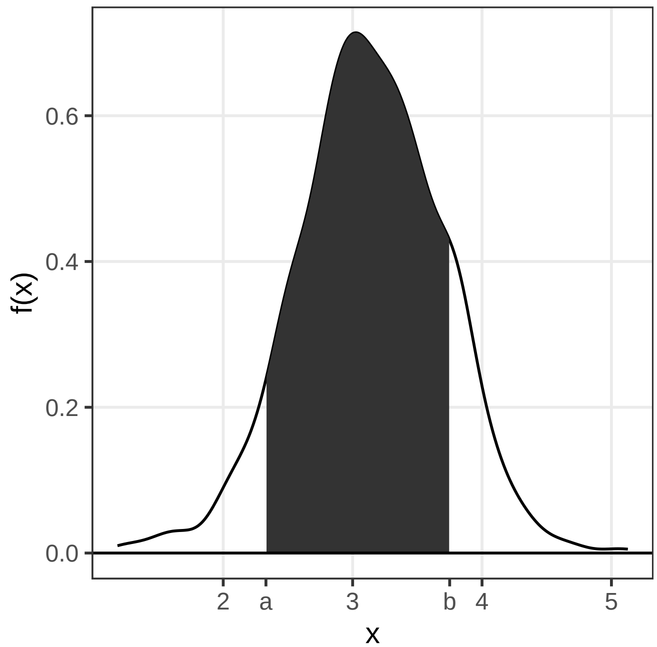


Figure 3.2: The probability $P(a \leq X \leq b)$ can be computed by computing the area under the probability density function between a and b.

$$P(a \leq X \leq b) = \int_a^b f(x)dx$$

The **cumulative distribution function**, cdf, sometimes called just the distribution function, $F(x)$, is defined as:

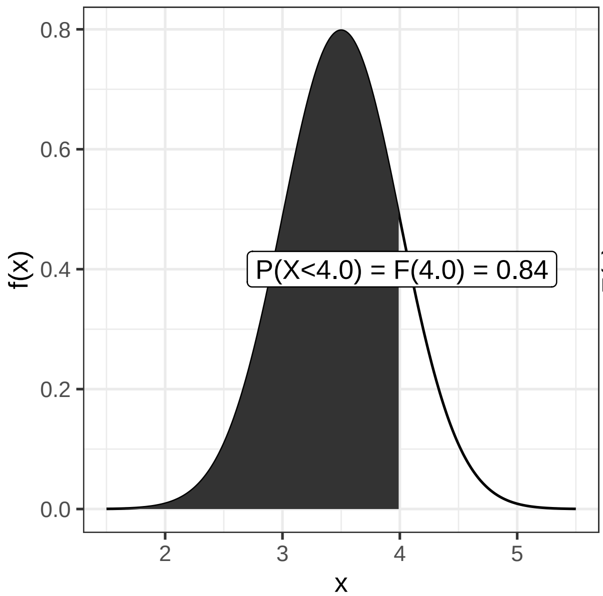
$$F(x) = P(X \leq x) = \int_{-\infty}^{x} f(t)dt$$

```
Warning in geom_point(aes(x = 4, y = pnorm(4, 3.5, 0.5))): All aesthetics have length 1, but
i Please consider using `annotate()` or provide this layer with data containing
  a single row.
```

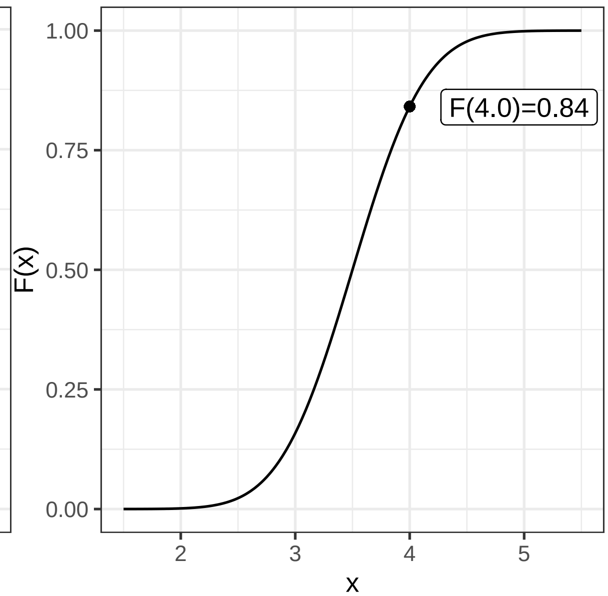
$$P(X \leq x) = F(x)$$

As we know that the total probability (over all x) is 1, we can conclude that

$$P(X > x) = 1 - F(x)$$



(a) PDF



(b) CDF

Figure 3.3: Probability density function (pdf) and cumulative distribution function (cdf).

and thus

$$P(a < X \leq b) = F(b) - F(a)$$

3.1 Parametric continuous distributions

Two important parameters of a distribution is the expected value, μ, that describe the distributions location and the variance, σ^2, that describe the spread.

The expected value, or population mean, is defined as;

$$E[X] = \mu = \int_{-\infty}^{\infty} x f(x) dx$$

We will learn more about the expected value and how to estimate a population mean from a sample later in the course.

The population variance is defined as the expected value of the squared distance from the population mean;

$$\sigma^2 = E[(X-\mu)^2] = \int_{-\infty}^{\infty} (x-\mu)^2 f(x) dx$$

The square root of the variance is the standard deviation, σ.

3.2 Normal distribution

The normal distribution (sometimes referred to as the Gaussian distribution) is a common bell-shaped probability distribution. Many continuous random variables can be described by the normal distribution or be approximated by the normal distribution.

The normal probability density function

$$f(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

describes the distribution of a normal random variable, X, with expected value μ and standard deviation σ, e and π are two common mathematical constants, $e \approx 2.71828$ and $\pi \approx 3.14159$.

The bell-shaped normal distributions is symmetric around μ and $f(x) \to 0$ as $x \to \infty$ and as $x \to -\infty$.

As $f(x)$ is well defined, values for the cumulative distribution function $F(x) = \int_{-\infty}^{x} f(x) dx$ can be computed.

If X is normally distributed with expected value μ and standard deviation σ we write:

$$X \sim N(\mu, \sigma)$$

Using transformation rules we can define Z, a random variable that is standard normal distributed (mean 0 and standard deviation 1).

$$Z = \frac{X-\mu}{\sigma},\ Z \sim N(0,1)$$

Values for the cumulative standard normal distribution, $F(z)$, are tabulated and easy to compute in R using the function `pnorm`.

Table 3.1: Cumulative distribution function for the standard normal distribution. The table gives F(z) = P(Z < z) for standard normal Z.

0	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09

0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
3.0	0.9987	0.9987	0.9987	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990	0.9990
3.1	0.9990	0.9991	0.9991	0.9991	0.9992	0.9992	0.9992	0.9992	0.9993	0.9993
3.2	0.9993	0.9993	0.9994	0.9994	0.9994	0.9994	0.9994	0.9995	0.9995	0.9995
3.3	0.9995	0.9995	0.9995	0.9996	0.9996	0.9996	0.9996	0.9996	0.9996	0.9997
3.4	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9998

Some values of particular interest:

$$F(1.64) = 0.95$$

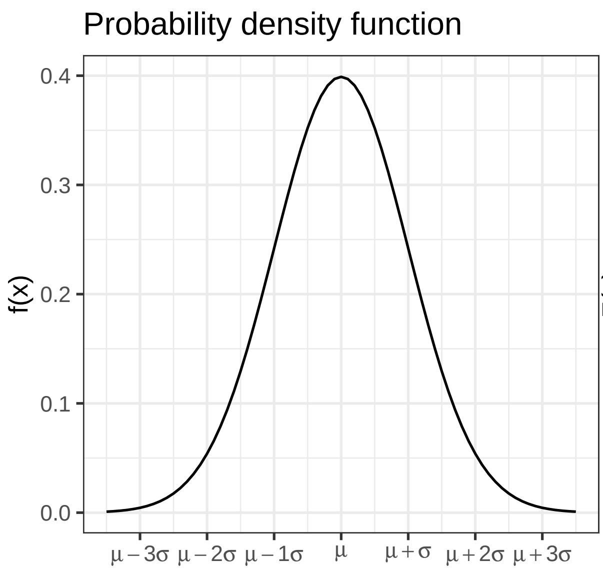


Figure 3.4: Normal probability density function and cumulative distribution functions.

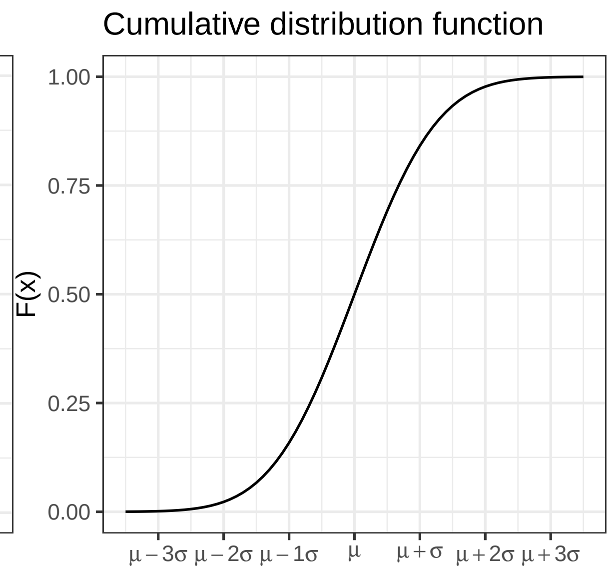


Figure 3.5: Normal probability density function and cumulative distribution functions.

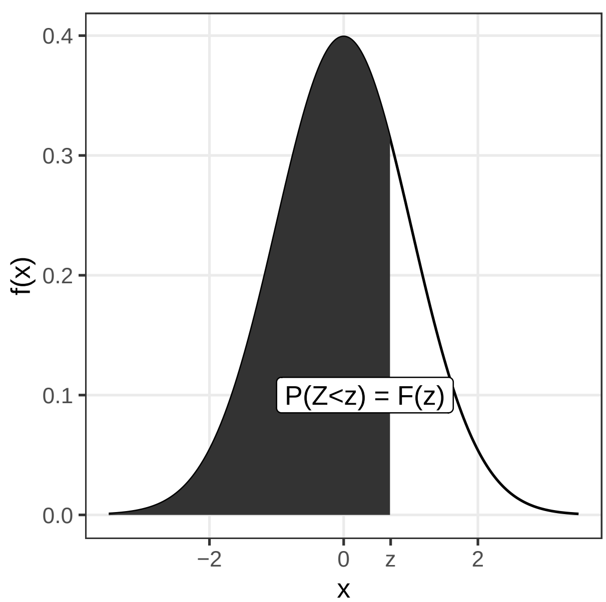


Figure 3.6: The shaded area under the curve is the tabulated value $P(Z \leq z) = F(z)$.

$$F(1.96) = 0.975$$

As the normal distribution is symmetric F(-z) = 1 - F(z)

$$F(-1.64) = 0.05$$

$$F(-1.96) = 0.025$$

$$P(-1.96 < Z < 1.96) = 0.95$$

3.2.1 Sum of two normal random variables

If $X \sim N(\mu_1, \sigma_1)$ and $Y \sim N(\mu_2, \sigma_2)$ are two independent normal random variables, then their sum is also a random variable:

$$X + Y \sim N(\mu_1 + \mu_2, \sqrt{\sigma_1^2 + \sigma_2^2})$$

and

$$X - Y \sim N(\mu_1 - \mu_2, \sqrt{\sigma_1^2 + \sigma_2^2})$$

This can be extended to the case with n independent and identically distributed random variables X_i $(i = 1 \dots n)$. If all X_i are normally distributed with mean μ and standard deviation σ, $X_i \in N(\mu, \sigma)$, then the sum of all n random variables will also be normally distributed with mean $n\mu$ and standard deviation $\sqrt{n}\sigma$.

3.3 Central limit theorem

Theorem 3.1 (CLT)**.** *The sum of n independent and equally distributed random variables is normally distributed, if n is large enough.*

As a result of the central limit theorem, the distribution of fractions or mean values of a sample follow the normal distribution, at least if the sample is large enough (a rule of thumb is that the sample size $n > 30$).

Example 3.1 (A skewed distribution)**.** A left skewed distribution has a heavier left tail than right tail. An example might be age at death of natural causes.

Randomly sample 3, 5, 10, 15, 20, 30 values and compute the mean value, m. Repeat many times to get the distribution of mean values.

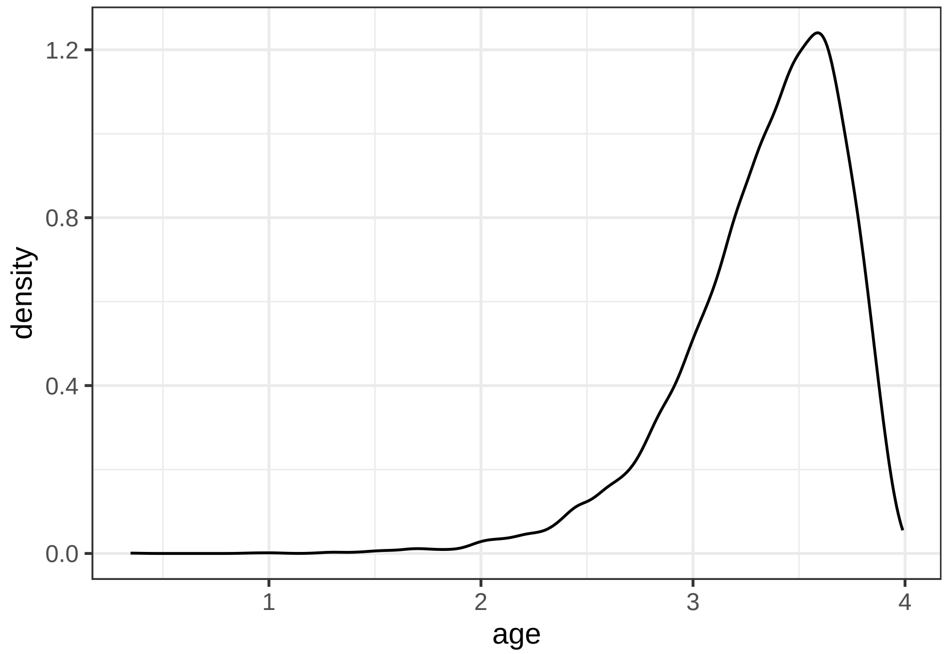


Figure 3.7: A left skewed distribution. Can for example show the distribution of age of a mouse who died of natural causes.

```
Warning: Returning more (or less) than 1 row per `summarise()` group was deprecated in
dplyr 1.1.0.
i Please use `reframe()` instead.
i When switching from `summarise()` to `reframe()`, remember that `reframe()`
  always returns an ungrouped data frame and adjust accordingly.
```

Note, mean is just the sum divided by the number of samples n.

3.4 χ^2-distribution

The random variable $Y = \sum_{i=1}^{n} X_i^2$ is χ^2 distributed with $n-1$ degrees of freedom, if X_i are independent identically distributed random variables $X_i \in N(0,1)$.

In short $Y \in \chi^2(n-1)$.

Example 3.2. The sample variance $S^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X})^2$ is such that $\frac{(n-1)S^2}{\sigma^2}$ is χ^2 distributed with $n-1$ degrees of freedom.

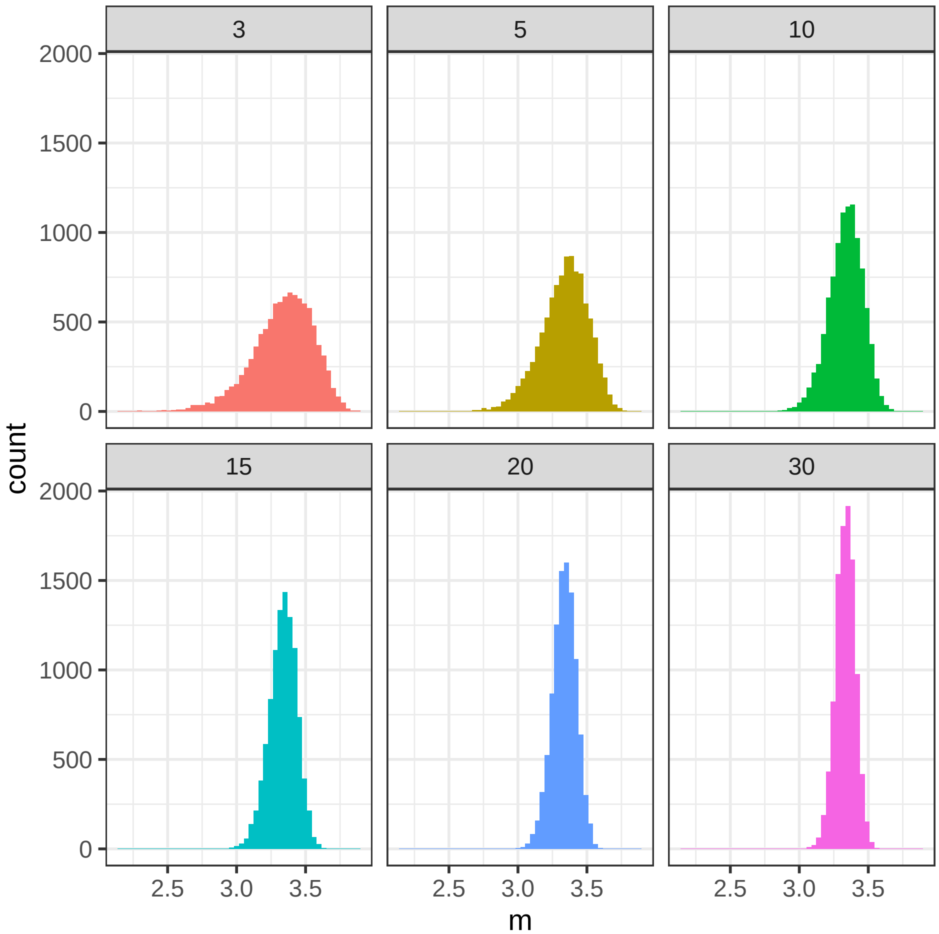


Figure 3.8: Distribution of sample means, where the means are computed based on random samples of sizes 3, 5, 10, 15, 20 and 30, respectively.

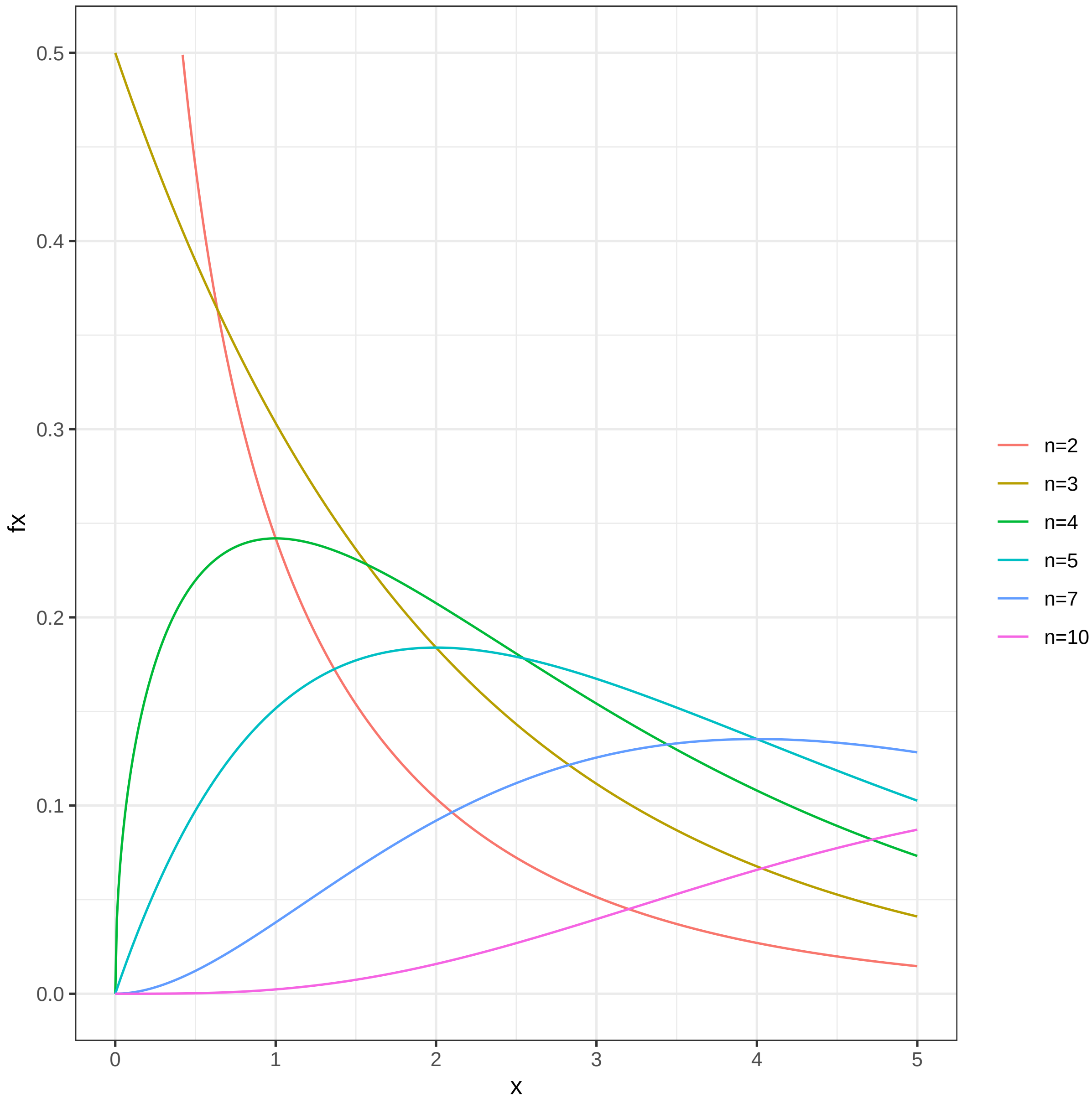


Figure 3.9: The χ^2-distribution.

3.5 F-distribution

The ratio of two χ^2-distributed variables divided by their degrees of freedom is F-distributed

Example 3.3. The ratio of two sample variances is F-distributed

3.6 t-distribution

The ratio of a normally distributed variable and a χ^2-distributed variable is t-distributed.

Example 3.4. The ratio between sample mean and sample variance is t-distributed.

3.7 Distributions in R

Probability density functions for the normal, t, χ^2 and F distributions can in R can be computed using functions `dnorm`, `dt`, `dchisq`, and `df`, respectively.

Cumulative distribution functions can be computed using `pnorm`, `pt`, `pchisq` and `pf`.

Also, functions for computing an x such that $P(X < x) = q$, where q is a probability of interest are available using `qnorm`, `qt`, `qchisq` and `qf`.

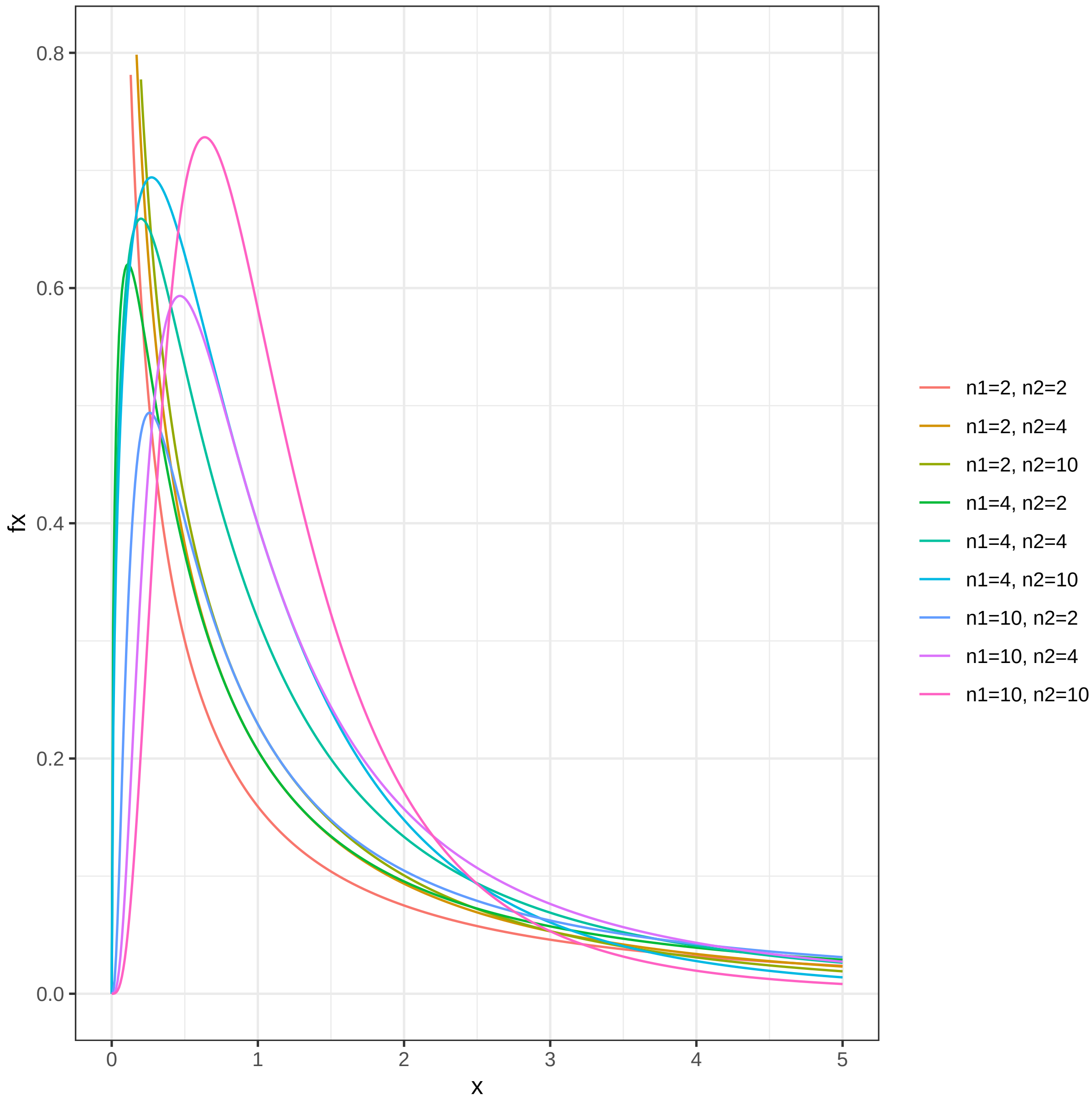


Figure 3.10: The F-distribution

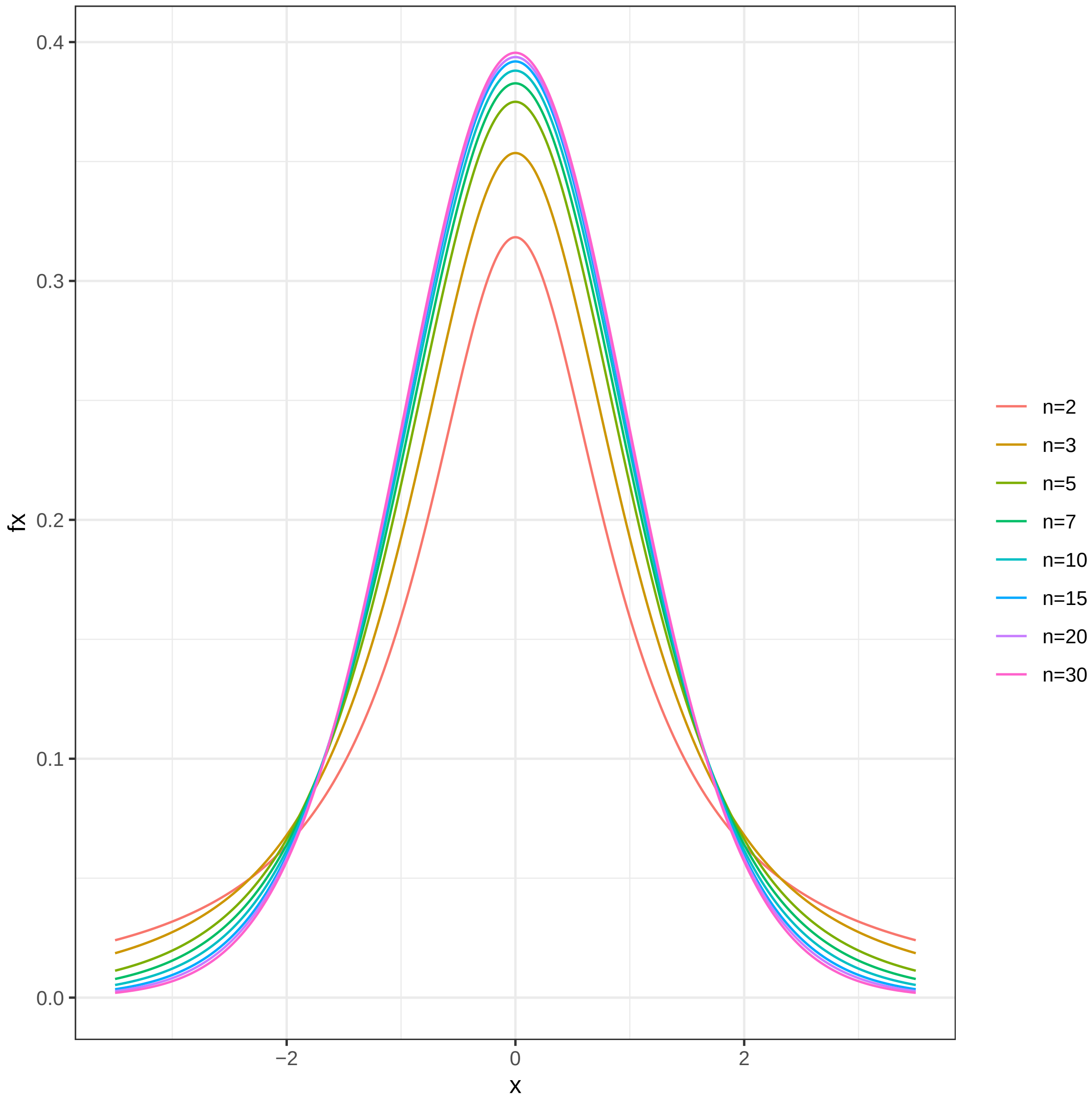


Figure 3.11: The t-distribution.

4 Sampling and experimental design

4.1 Random sampling

In many (most) experiments it is not feasible (or even possible) to examine the entire population. Instead a random sample is studied.

A **random sample** is a random subset of individuals from a population.

There are different techniques for performing random sampling, two common techniques are **simple random sampling** and **stratified random sampling**.

A **simple random sample** is a random subset of individuals from a population, where every individual has the same probability of being choosen. Simple random sampling can be performed using the urn model. If every individual in the population is represented by one ball, a simple random sample of size n can be achieved by drawing n balls from the urn, without replacement.

In **stratified random sampling** the population is first divided into subpopulations based on important attributes, e.g. sex (male/female), age (young/middle aged/old) or BMI (underweight/normal weight/overweight/obese). Simple random sampling is then performed within each subpopulation.

4.2 Principles of experimental design

When you plan your experiment and assign experimental units to treatment/control group (or similar), it is important to consider extraneous variables, i.e. variables that are not your main interest but that might affect the studied experimental outcome or the variable of interest. These variables can be properties of the experimental units such as age, sex etc. but also variables introduced in the experiment, like batch, experiment date, laboratory personell etc.

Fundamental to experimental design are the three principles; **replication**, **randomization** and **blocking**.

Replication. Replication is the repetition of the same experiment, with the same conditions. **Biological replicates** are measurements of different biological units under the same conditions, whereas **technical replicates** are repeated measurements of the same biological unit under the same conditions.

Randomization. Experimental units are not identical, hence by assigning experimental units to treatment/control at random we can avoid unnecessary bias. It is also important to perform the measurements in random order.

Blocking. Blocking is grouping experimental units into blocks consisting of units that are similar to one another and assigning units within a block to treatment/control at random. Blocking reduces known but irrelevant sources of variation between units and thus allows greater precision in the estimation of the source of variation under study.

The experimental units can be grouped into blocks according to their properies (e.g. age, sex, etc). Units within a block will be more similar than between blocks. By assigning units within a block to treatment/control at random, the variation due to differences between blocks (that are not relevant to the studied outcome) can be reduced.

In many retrospective studies it is not possible to assign patients into treatment/control or sick/healthy. Experimental design is still important for controling sources of variation introduced during the experiment.

Block what you can; randomize what you cannot.

4.3 Sample properties

Summary statistics can be computed for a sample, such as the sum, proportion, mean and variance.

Notation: A random sample $x_1, x_2, \ldots, x_n$ from a distribution, D, consists of n observations of the independent random variables $X_1, X_2, \ldots, X_n$ all with the probability distribution D.

4.3.1 Sample proportion

The proportion of a population with a particular property, the **population proportion**, is denoted π.

The number of individuals with the property in a simple random sample of size n is a random variable X. The proportion of individuals in a sample with the property is also a random variable;

$$P = \frac{X}{n}$$

with expected value

$$E[P] = \frac{E[X]}{n} = \frac{n\pi}{n} = \pi$$

.

The sample proportion, p, is said to be an **unbiased** estimate of the **population proportion**, as it's expected value is the population proportion π.

4.3.2 Sample mean

For a particular sample of size n; $x_1, \dots, x_n$, the sample mean is denoted $m = \bar{x}$. The sample mean is calculated as;

$$m = \bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i.$$

Note that the mean of n independent identically distributed random variables, X_i, is itself a random variable;

$$\bar{X} = \frac{1}{n}\sum_{i=1}^{n} X_i.$$

If X_i are normaly distributed $X_i \sim N(\mu, \sigma)$, then $\bar{X} \sim N\left(\mu, \frac{\sigma}{\sqrt{n}}\right)$.

When we only have a sample of size n, the sample mean m is our best estimate of the population mean. It is possible to show that the sample mean is an **unbiased** estimate of the population mean, i.e. the average (over many size n samples) of the sample mean is μ.

$$E[\bar{X}] = E\left[\frac{1}{n}\sum_{i=1}^{n} X_i\right] = \frac{1}{n}\sum_{i=1}^{n} E[X_i] = \frac{1}{n} n\mu = E[X] = \mu$$

4.3.3 Sample variance

The sample variance is computed as;

$$s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - m)^2.$$

The sample variance is an unbiased estimate of the population variance.

4.3.4 Sampling distribution

A sampling distribution is a probability distribution of a sample property. A sampling distribution is obtained by sampling many times from the studied population.

4.3.5 Standard error

Eventhough the sample can be used to calculate unbiased estimates of the population property, the sample estimate will not be perfect. The standard deviation of the sampling distribution is called the **standard error**, SE.

For the sample mean, $\bar{X}$, the variance is

$$E[(\bar{X}-\mu)] = var(\bar{X}) = var(\frac{1}{n}\sum_i X_i) = \frac{1}{n^2}\sum_i var(X_i) = \frac{1}{n^2} n var(X) = \frac{\sigma^2}{n}$$

The standard error of the mean is thus;

$$SEM = \frac{\sigma}{\sqrt{n}}$$

Replacing σ with the sample standard deviation, s, we get an estimate of the standard error of the mean;

$$SEM \approx \frac{s}{\sqrt{n}}$$

Exercises: Continuous random variables

Parametric continuous distribution

Exercise 4.1 (The normal table)**.** Let $Z \sim N(0,1)$ be a standard normal random variable, and compute;

a. $P(Z < 1.64)$
b. $P(Z > -1.64)$
c. $P(-1.96 < Z)$
d. $P(Z < 2.36)$
e. An a such that $P(Z < a) = 0.95$
f. A b such that $P(Z > b) = 0.975$

Note, this exercise can be solved using the standard normal table or using the R functions `pnorm`and `qnorm`.

Solution

a. $P(Z < 1.64) = P(Z \leq 1.64) = \text{[from table]} = 0.9495$
b. $P(Z > -1.64) = \text{[symmetry]} = P(Z < 1.64) = 0.9495$
c. $P(-1.96 < Z) = \text{[symmetry]} = P(Z < 1.96) = \text{[from table]} = 0.975$
d. $P(Z < 2.36) = \text{[from table]} = 0.9909$
e. Standard normal table gives that $a = 1.64$
f. $P(Z > b) = 0.975$, symmetry gives that $P(Z < -b) = 0.975$. Look-up in the standard normal table gives that $-b = 1.96$, hence $b = -1.96$.

Exercise 4.2 (Exercise in standardization/transformation)**.** If $X \sim N(3,2)$, compute the probabilities

a. $P(X < 5)$
b. $P(3 < X < 5)$
c. $P(X \geq 7)$

ℹ Solution

a. $P(X<5) = P\left(\frac{X-3}{2} < \frac{5-3}{2}\right) = P(Z<1) = [\text{from table}] = 0.8413$
b. $P(3<X<5) = P\left(\frac{3-3}{2} < \frac{X-3}{2} < \frac{5-3}{2}\right) = P(0<Z<1) = P(Z<1) - P(Z<0) = 0.8413 - 0.5000 = 0.3413$
c. $P(X \geq 7) = P\left(\frac{X-3}{2} \geq \frac{7-3}{2}\right) = P(Z \geq 2) = 1 - P(Z<2) = 1 - 0.9772 = 0.0228$

Exercise 4.3 (Hemoglobin)**.** The hemoglobin (Hb) value in a male population is normally distributed with mean 188 g/L and standard deviation 14 g/L.

a) Men with Hb below 158 g/L are considered anemic. What is the probability of a random man being anemic?
b) When randomly selecting 10 men from the population, what is the probability that none of them are anemic?

ℹ Solution

a) $P(Hb < 158) = P(Z < \frac{158-188}{14}) = P(Z < -2.14) = [table] = 0.016$ or use R

```
pnorm(158, mean=188, sd=14)
```

```
[1] 0.01606229
```

b) Probability of one man not being anemic; $1 - 0.016 = 0.984$. Probability of 10 selected men not being anemic (binomial distribution) $0.984^{10} = 0.85$

Exercise 4.4 (Pill)**.** A drug company is producing a pill, with on average 12 mg of active substance. The amount of active substance is normally distributed with mean 12 mg and standard deviation 0.5 mg, if the production is without problems. Sometimes there is a problem with the production and the amount of active substance will be too high or too low, in which case the pill has to be discarded. What should the upper and lower critical values (limits for when a pill is acceptable) be in order not to discard more than 1/20 pills from a problem free production?

ℹ Solution

$H_0 : \mu = 12$ $H_1 : \mu \neq 12$
Search x_{low} and x_{up} such that $P(x_{low} < X < x_{up}) = 0.05$
From table we know that $P(-1.96 < Z < 1.96) = 0.05$
$Z = \frac{X-\mu_0}{\sigma_0} = \frac{X-12}{0.5}$, hence
$-1.96 < Z < 1.96 \iff -1.96 < \frac{X-12}{0.5} < 1.96 \iff 12 - 1.96 * 0.5 < X < 12 + 1.96 *$

$0.5 \iff 11.02 < X < 12.98$
The lower and upper critical values of active substance should be 11.02 and 12.98 mg.

Random sample

Exercise 4.5 (Exercise in distribution of sample mean)**.** The total cholesterol in population (mg/dL) is normally distributed with $\mu = 202$ and $\sigma = 40$.

a. How is the sample mean of a sample of 4 persons distributed?
b. What is the probability to see a sample mean of 260 mg/dL or higher?
c. Is there reason to believe that the four persons with mean 260 mg/dL were sampled from another population with higher population mean?

Solution

a.
$$\bar{X} = \frac{1}{4}\sum_{i=1}^{4} X_i X_i \sim N(\mu, \sigma)\bar{X} \sim N\left(\mu, \frac{\sigma}{\sqrt{n}}\right) = N(202, 20)$$
b. 0.0019
c. Yes

Exercise 4.6 (Amount of active substance)**.** The amount of active substance in a pill is stated by the manufacturer to be normally distributed with mean 12 mg and standard deviation 0.5 mg. You take a sample of five pills and measure the amount of active substance to; 13.0, 12.3, 12.6, 12.5, 12.7 mg.

[Note: a-c were already computed in the descriptive statistics session.]

a) Compute the sample mean
b) Compute the sample variance
c) Compute the sample standard deviation
d) compute the standard error of mean, SEM.
e) If the manufacturers claim is correct, what is the probability to see a sample mean as high as in a) or higher?

Answer

a. 12.62
b. 0.067
c. 0.26
d. 0.22
e. 0.0028

f. Sample mean

Solution

```
x <- c(13.0, 12.3, 12.6, 12.5, 12.7)
n <- length(x)
(m <- sum(x)/n)
```

```
[1] 12.62
```

b. Sample variance

```
[1] 0.067
```

c. Sample standard deviation

```
[1] 0.2588436
```

d. Standard error of mean, SEM

```
[1] 0.2236068
```

Note, here the known standard deviation, $\sigma = 0.5$ is used.

References